

JANUARY 27, 2020

1. *At 25°C, an aqueous solution containing 35.0 wt% H_2SO_4 has a specific gravity of 1.2563. A quantity of the 35% solution is needed that contains 195.5 kg of H_2SO_4 .*

a) Calculate the required volume (L) of the solution using the given specific gravity.

b) Estimate the percentage error that would have resulted if pure-component specific gravities of H_2SO_4 (SG=1.8255) and water has been used for the calculation instead of the given specific gravity of the mixture.

2. You are trying to decide which of two automobiles to buy. The first is American-made, costs \$14,500, and has a rated gasoline mileage of 28 miles/gal. The second car is of European manufacture, costs \$21,700, and has a rated mileage of 19 km/L. If the cost of gasoline is \$1.25/gal and if the cars actually deliver their rated mileage, estimate how many miles you would have to drive for the lower fuel consumption of the second car to compensate for the higher cost of this car?

Useful information: 1 gal = 3.785 L; 1 mi = 1609.34 m